ACTIVITY-4

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import skew, kurtosis
np.random.seed(42)
normal_data = np.random.normal(loc=50, scale=5, size=1000)
skewed_data = np.random.exponential(scale=10, size=1000)
plt.figure()
sns.histplot(normal_data, kde=True)
plt.title("Normal Distribution")
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.show()
plt.figure()
sns.histplot(skewed_data, kde=True)
plt.title("Exponential (Right-Skewed) Distribution")
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.show()
print("Skewness:")
print("Normal Data:", skew(normal_data))
print("Skewed Data:", skew(skewed_data))
print("\nKurtosis (Fisher's Excess Kurtosis):")
print("Normal Data:", kurtosis(normal_data))
print("Skewed Data:", kurtosis(skewed_data))
def calculate_z_scores(data):
    mean = np.mean(data)
    std = np.std(data)
```

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FDS.ipynb

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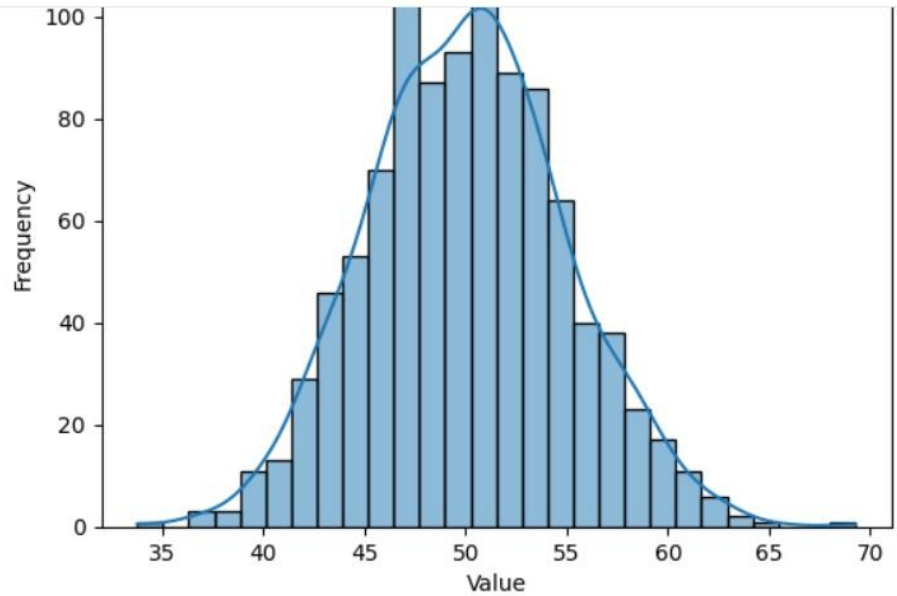
```
    std = np.std(data)
    return (data - mean) / std
z_scores = calculate_z_scores(skewed_data)
outliers = skewed_data[np.abs(z_scores) > 3]
print("\nNumber of Outliers in Skewed Data:", len(outliers))
```



Exponential (Right-Skewed) Distribution

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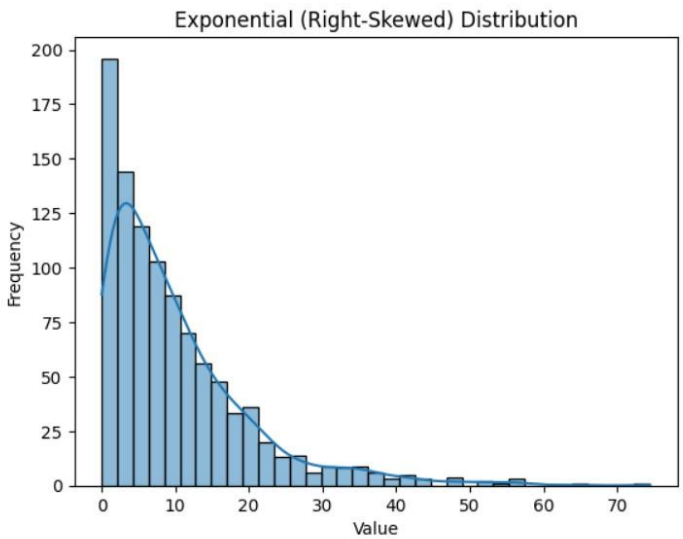
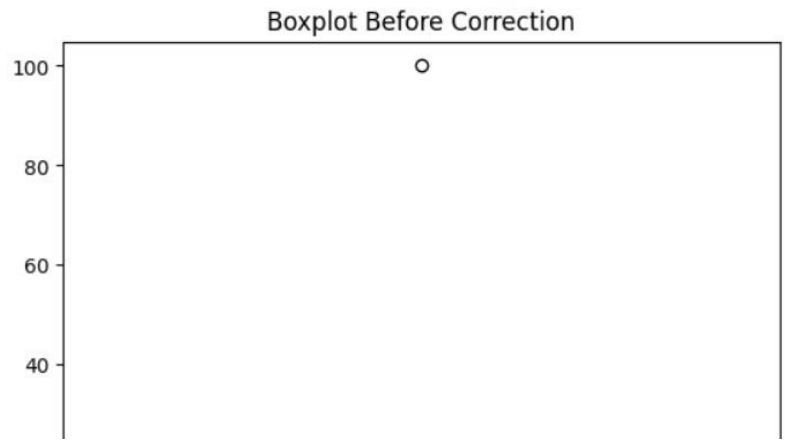


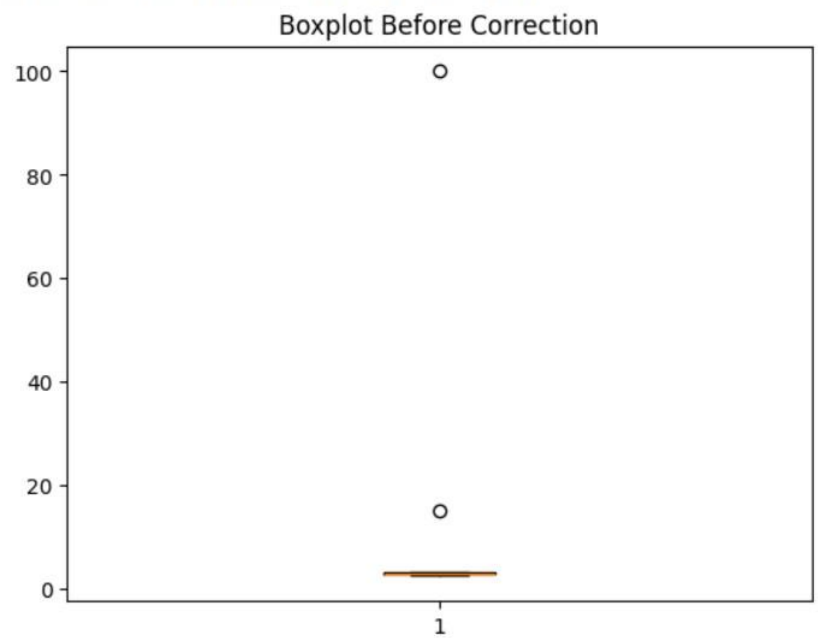

```
Skewness:
Normal Data: 0.11680083110533704
Skewed Data: 1.9808119458203122

Kurtosis (Fisher's Excess Kurtosis):
```

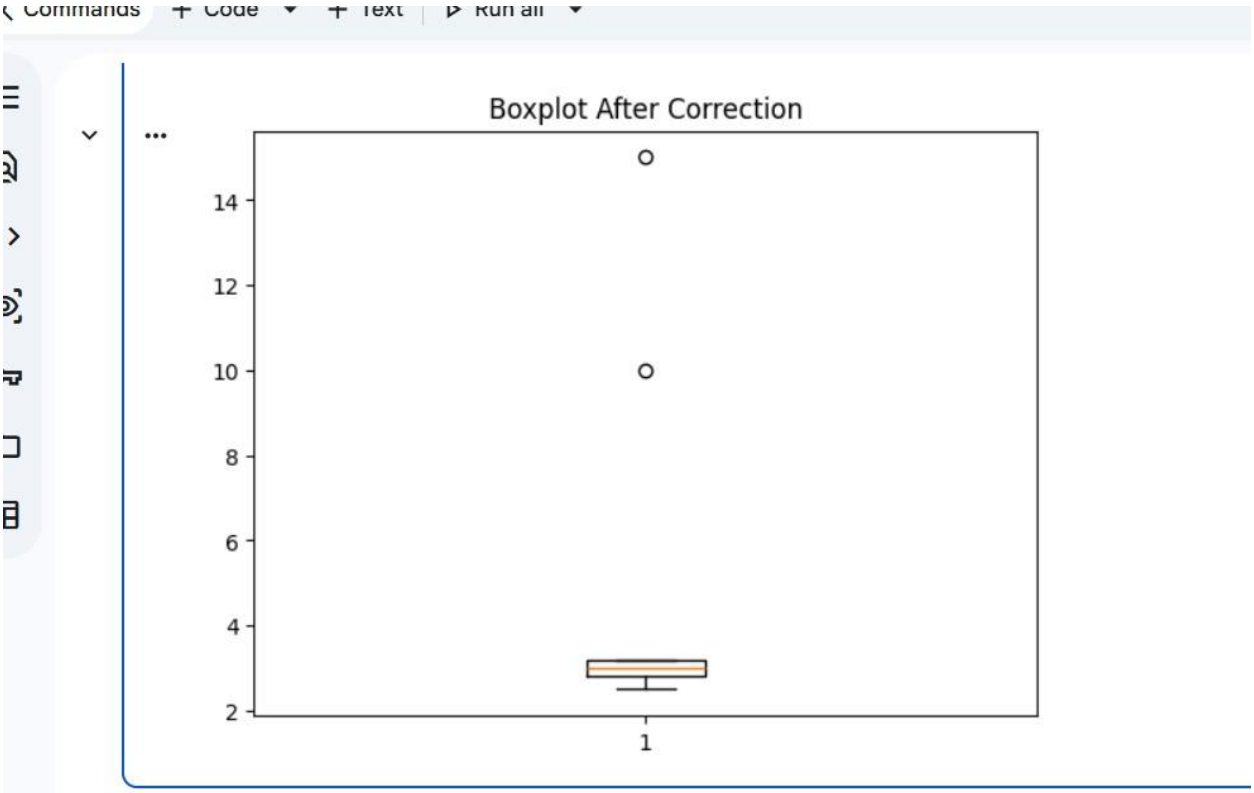
```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import zscore
prices = np.array([2.5, 3.0, 2.8, 3.2, 15.0, 2.9, 3.1, 2.7, 100.0])
print("Original Data:")
print(prices)
z_scores = zscore(prices)
outliers = prices[np.abs(z_scores) > 3]
print("\nZ-Scores:")
print(z_scores)
print("\nOutliers Detected (|Z| > 3):")
print(outliers)
prices_corrected = np.where(prices == 100.0, 10.0, prices)
print("\nCorrected Data:")
print(prices_corrected)
plt.figure()
plt.boxplot(prices)
plt.title("Boxplot Before Correction")
plt.show()
plt.figure()
plt.boxplot(prices_corrected)
plt.title("Boxplot After Correction")
plt.show()
```

```
Original Data:
[  2.5   3.    2.8   3.2  15.    2.9   3.1   2.7 100. ]

Z-Scores:
[-4.13529340e-01 -3.97017520e-01 -4.03622248e-01 -3.90412793e-01
 -7.33858633e-04 -4.00319884e-01 -3.93715157e-01 -4.06924612e-01
  2.80627541e+00]

Outliers Detected (|Z| > 3):
[]

Corrected Data:
[ 2.5  3.   2.8  3.2 15.   2.9  3.1  2.7 10. ]
```




```
Corrected Data:
[ 2.5  3.   2.8  3.2 15.   2.9  3.1  2.7 10. ]
```




Boxplot After Correction
14
12
10
8
6
4
2
1